

Notes on Krylov complexity talk-EUPAC

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1 Physical Intuition of Operator growth

Lets start with a very simple, highly localised operator $\mathcal{O}_0$. This operator could be imagine as a single Pauli spin operator acting on a single site of a lattice. As we move with time, the Heisenberg mechanism gives us

$$\mathcal{O}(t) = e^{iHt}\mathcal{O}_0e^{-iHt} \quad (1)$$

this is the Heisenberg picture of the operator at time t . Our operator at time t can be expanded into an infinite series of nested commutators using the Baker-Campbell-Hausdorff expansion.

$$\mathcal{O}(t) = \mathcal{O}_0 + it[H, \mathcal{O}_0] - \frac{t^2}{2}[H, [H, \mathcal{O}_0]] + \dots \quad (2)$$

At every term the $\mathcal{O}_0$ is act on by the commutator $[H, \cdot]$, the operator "grows". It starts to coupled to the next neighbor and the the next and the next and so on... eventually our operator evolved into this massive, highly entangled superposition of operators which spans the entire system. "Complexity" quantify the "size" of this monster, we have created, at time t .

2 Circuit complexity

I am going to focus on Nelson's circuit complexity [1] as this was the bedrock of the entire complexity as geometry movement that eventually gives rise to Krylov complexity.

The story starts around 2005 with Michael Nielsen. He wanted a way to lower bound the size of quantum circuits, and he realized that quantum computation could be recast entirely as a problem in Riemannian geometry. In quantum computing, an algorithm is a sequence of discrete unitary gates, like CNOT, Hadamard, which acts on a set of N qubits. The idea was to replace these discrete gates by a continuous Hamiltonian evolution for an easier analytical way. Instead of applying gates step-by-step, imagine a time-dependent Hamiltonian $H(t)$ that drives the system from the initial state (say, $|0\rangle$) to the target state $|\psi\rangle$, or from the identity operator to a target unitary U . The evolution operator is now a path in the manifold of $SU(2^N)$ matrices

$$U(s) = \mathcal{P} \exp \left(-i \int_0^s H(s') ds' \right) \quad (3)$$

where $s \in [0, 1]$ parameterizes the path, $U(0) = \mathbb{I}$, and $U(1) = U_{\text{target}}$. The "velocity" along this path is just the Hamiltonian:

$$H(s) = i \left(\frac{dU(s)}{ds} \right) U^\dagger(s) \quad (4)$$

Not all directions in $SU(2^N)$ are created equal. In a real quantum computer, a 2-qubit gate is easy, but a 15-qubit gate is incredibly hard. Nielsen expanded the instantaneous Hamiltonian in a basis of generalized Pauli strings (which act as the generators of the algebra)

$$H(s) = \sum_I Y^I(s) \mathcal{O}_I \quad (5)$$

Here, $\mathcal{O}_I$ are combinations of Pauli matrices (e.g., $X_1 \otimes Z_2 \otimes \mathbb{I} \dots$), and $Y^I(s)$ are the control coefficients, these are just real numbers that tell you how much of the Pauli string $\mathcal{O}_I$ you are applying at step s .

Lets define the geometry, we consider a right invariant Riemannian metric

$$ds^2 = \sum_{I,J} \mathcal{G}_{IJ} Y^I(s) Y^J(s) ds^2 \quad (6)$$

here we introduce $\mathcal{G}_{IJ}$ which is an arbitrary weights that we assign (this weight factor makes the manifold curve). Nielson set out to assign massive penalty weights to complex, highly non-local operators (like a 10-qubit coupling) so the geometry forces the path to stay along directions composed of simple 1- and 2-qubit gates.

And now our problem reduces to finding the shortest path. The circuit complexity $\mathcal{C}$ is the length of this geodesic

$$\mathcal{C} = \min_{H(s)} \int_0^1 \sqrt{\sum_I \mathcal{G}_{II} (Y^I(s))^2} ds \quad (7)$$

2.1 $SU(2^N)$ group

All quantum circuits live in this mathematical space. A single qubit exists in a 2 dimensional complex vector space $\mathbb{C}^2$, spanned by $|0\rangle$ and $|1\rangle$. If our system has N qubits, the total space is the tensor product of all individual qubit spaces

$$\mathbb{C}^2 \otimes \mathbb{C}^2 \dots \otimes \mathbb{C}^2 = \mathbb{C}^{2^N} \quad (8)$$

Hence, our vector describing our N -qubit has 2^N complex entries. So, any operation that seeks to transform the states is represented by $2^N \times 2^N$ matrix.

All transformation in quantum mechanics must be unitary, satisfying $U^\dagger U = \mathbb{I}$, which makes it a $U(2^N)$. Furthermore, physical observation must be invariant under global phase shift ($e^{i\theta}|\psi\rangle$ is physically identical to $|\psi\rangle$), which restrict us to matrices that have a determinant of exactly 1. This gives us the Special Unitary group of degree 2^N , denoted as $SU(2^N)$.

$SU(2^N)$ is a Lie group, which means it possesses both the algebraic structure of a mathematical group and the continuous, smooth geometry of a differentiable manifold. A point on the manifold represents a specific, complete quantum circuit (a specific unitary matrix). Identity, $\mathbb{I}$ represents doing absolutely nothing to your qubits. A path on the manifold represents the continuous application of a Hamiltonian over time, gradually building up your circuit.

For the set of special unitary matrices to form a group, they must satisfy four algebraic properties.

- **Closure :** If you have two valid quantum operations, U_1 and U_2 , applying them sequentially ($U_2 U_1$) results in a new matrix that is also in $SU(2^N)$. You never “leave” the manifold by combining operations.
- **Associativity :** When combining multiple operations, $(U_3 U_2) U_1 = U_3 (U_2 U_1)$. The way you group the intermediate gates doesn’t matter, as long as the temporal sequence remains the same.
- **Identity:** There must exist an element that leaves everything unchanged. This is the Identity matrix $\mathbb{I}$. Geometrically, this is the “origin” of our manifold where all complexity paths begin.
- **Invertibility :** This is the most crucial physical constraint. Every operation must be perfectly reversible. For every unitary matrix U , there exists an inverse U^{-1} (which is exactly its conjugate transpose, $U^\dagger$) such that $U U^\dagger = \mathbb{I}$. If you run a quantum circuit forward, you can always run it backward to return to the identity.

Because of the algebraic structure, the manifold looks exactly the same from every single point. This allows us to use Right-Invariance. If you want to find the shortest path (the complexity) between two highly complicated quantum circuits U_A and U_B , you don’t actually need to solve the geometry out in the middle of nowhere. Because of the group rules, you can multiply both points by $U_A^\dagger$. U_A becomes $\mathbb{I}$ and U_B becomes $U_B U_A^\dagger$ which is the new target unitary. The group algebra allows Nielsen to shift any complexity calculation back to the origin, calculating how hard it is to build the relative difference between the two operators starting from zero.

2.2 Why Right invariance?

In geometry, an "invariant" metric means the distance between two points doesn't change when you shift or translate the entire space. For the $SU(2^N)$ manifold of quantum circuits, "translating" means matrix multiplication. Because matrix multiplication isn't commutative ($AB \neq BA$), we have two different types of translations: multiplying from the right, and multiplying from the left.

Let us consider two unitary metrics U_A and U_B , and the distance is define as $d(U_A, U_B)$. We say a metric is right invariant if the distance does not change when we multiply an unitary matrix W on the right side

$$d(U_A, U_B) = d(U_A W, U_B W) \quad (9)$$

Right multiplying W represent changing the initial setup or the history of the circuit before U is applied (W happen first). A right-invariant metric means the cost of applying a gate right now is completely independent of the circuit's history. If you apply a CNOT gate, it costs exactly "1 unit of complexity" regardless of whether you are applying it to a pristine $|00\rangle$ state or a massively entangled, scrambled state. The instantaneous cost depends only on the gates you are actively applying, not on the unitary you have already built. We can see this in the definition of the instantaneous Hamiltonian (the "velocity" of the circuit). Notice this is just Schrodinger equation

$$H(s) = i \frac{dU(s)}{ds} U^\dagger(s)$$

If we shift the whole path by right-multiplying a constant W ($U \rightarrow UW$), the new velocity is:

$$H_{new} = i \frac{d(UW)}{ds} (UW)^\dagger = i \frac{dU}{ds} W W^\dagger U^\dagger = i \frac{dU}{ds} U^\dagger = H(s)$$

The W perfectly cancels out. The Hamiltonian and the complexity cost is unchanged.

Left-multiplying by V ($U \rightarrow VU$) means applying V after U . If we check the Hamiltonian for a left-translation, the V does not cancel out:

$$H_{new} = i \frac{d(VU)}{ds} (VU)^\dagger = V \left(i \frac{dU}{ds} U^\dagger \right) V^\dagger = V H(s) V^\dagger$$

The Hamiltonian has been conjugated by V . Physically, this means your simple 2-qubit gate has been scrambled into a different basis. Moreover, a bi-invariant metric (both left and right invariant) would treat all directions in the $SU(2^N)$ manifold equally. It would say a 15-qubit gate costs the exact same as a 1-qubit gate. By breaking left-invariance, Nielsen introduces the "penalty factors" $\mathcal{G}_{IJ}$ that warp the geometry, making non-local gates incredibly expensive and forcing the geodesic path to use simple, local gates.

2.3 An Example

In order to avoid all the tensor products, lets consider a 1-qubit system. The manifold is $SU(2)$ and our generators are the three Pauli matrices $\sigma_x, \sigma_y, \sigma_z$.

Now lets impose some penalty of our choice (we will assign the $\mathcal{G}_{IJ}$ values). For moving in the X or Y direction, $\mathcal{G}_{XX} = \mathcal{G}_{YY} = 1$ and for moving on the Z direction $\mathcal{G}_{ZZ} = k^2$ where $k \gg 1$. Let say our target unitary is

$$U_{\text{target}} = \exp \left(-i \frac{\theta}{2} \sigma_z \right)$$

One way to go about this problem is to just rotate abot the Z -axis by an angle θ , drives straight to the result. But we shall see that this "naive" approach will cost us a lot. Using eq 7, we can calculate

$$\mathcal{C}_{\text{direct}} = \int_0^1 \sqrt{\mathcal{G}_{zz} \left(\frac{\theta}{2} \right)^2} ds = k \frac{\theta}{2} \quad (10)$$

Now, let us take a more "cheaper" path. Instead of rotation about Z -axis, consider rotation about X -axis by $-\frac{\pi}{2}$ than rotate about Y -axis by θ followed by rotation about X -axis by $\frac{\pi}{2}$. This is the standard Euler angle decomposition. Our cost is now

$$C_{\text{new}} = 1 \cdot \left(\frac{\pi}{4}\right) + 1 \cdot \left(\frac{\theta}{2}\right) + 1 \cdot \left(\frac{\pi}{4}\right) = \frac{\pi}{2} + \frac{\theta}{2} \quad (11)$$

By assigning a massive cost to σ_z , the Riemannian geometry naturally forced the shortest path to bend away from the Z -axis. The geodesic equation mathematically "discovered" the optimal way, although I have not shown that the latter is the geodesic, to compile an expensive Z -gate into a sequence of cheap X and Y gates.

2.4 Why some gates are "cheap" and others "costly"??

To perform a single-qubit gate (like an X or Y rotation), we just fire a specifically tuned microwave laser pulse at a specific atom or circuit. It is easy, fast, and relatively error-free. For a 2-qubit gate (like a CNOT), you have to physically couple two adjacent qubits, entangle them, and precisely control their interaction. This is much harder and prone to noise. For a much larger qubit unitary operation, say a 50-qubit unitary operation in a single step, you would need to perfectly couple 50 qubits simultaneously with a single, infinitely complex laser pulse. In the real world, this is physically impossible; we assign a high cost to such unitary operations. If we didn't make non-local gates mathematically expensive, the shortest path on the manifold to a target state would just be a single, massive straight line using an N -qubit Pauli string. The "complexity" would be $\mathcal{O}(1)$, which completely defeats the purpose of measuring how hard the circuit is to build.

This is a good time to pivot our discussion to Krylov complexity. All the above calculations are sweet and fine until we start asking what a 1-qubit gate is in (say) an AdS black hole. In such cases, we do not have the luxury of a well-defined meaning of locality, hence the metric penalties, $\mathcal{G}_{IJ}$, become completely arbitrary. Well, this is bad news for us. We need something that does not rely on external inputs.

3 Krylov Complexity

We have discuss the form an operator takes at certain time t

$$\mathcal{O}(t) = \mathcal{O}_0 + it[H, \mathcal{O}_0] - \frac{t^2}{2}[H, [H, \mathcal{O}_0]] + \dots$$

Lets clean this up a bit by defining the Liouvillian superoperator $\mathcal{L} \equiv [H, \cdot]$

$$\mathcal{O}(t) = \sum_{n=0} \frac{(it)^n}{n!} \mathcal{L}^n \mathcal{O}_0 \quad (12)$$

Notice our operator at all times t is entirely constrained to a subspace of the operator space. The minimal subspace that spans the entire time evolution of our operator is called **Krylov space** [2].

$$\mathcal{H}_{\mathcal{O}} = \text{span}\{\mathcal{L}^n \mathcal{O}_0\}_{n=0}^{\infty} = \text{span}\{\mathcal{O}_0, \mathcal{L}\mathcal{O}_0, \mathcal{L}^2\mathcal{O}_0, \dots\} \quad (13)$$

The total number of linearly independent elements in this set is called the dimension of the Krylov space K .

The full operator Hilbert space is a vast, multidimensional volume. But the time-evolved operator $\mathcal{O}(t)$ is strictly confined to the Krylov subspace. By restricting our analysis to $\mathcal{H}_{\mathcal{O}}$, we reduce the dimensionality of the problem. We are ignoring all the operators that our specific $\mathcal{O}(t)$ will never evolve into.

3.1 Defining an Inner product

To define an inner product, let me first define the expectation value of our operator. But before we do that we must asked ourselves What is the system background?? For example the expectation value of an operator $\mathcal{O}$ in a thermal ensamble is

$$\langle \mathcal{O} \rangle_\beta = \text{Tr}(\rho \mathcal{O}) = \frac{1}{Z} \text{Tr}(e^{-\beta H} \mathcal{O}) \quad (14)$$

where $\rho = \frac{e^{-\beta H}}{Z}$ and $\beta = \frac{1}{k_B T}$. We our temperature goes to infinity $T \rightarrow \infty$ ($\beta \rightarrow 0$), the thermal weight $e^{-\beta H}$ becomes and identity matrix $\mathbb{I}$. Hence, the partition function Z simply becomes the dimension of the Hilbert space D ¹

For an infinite temperature system, we use the Frobenius inner product, normalised by the Hilbert space dimension D

$$(\mathcal{A}|\mathcal{B}) = \frac{1}{D} \text{Tr}(\mathcal{A}^\dagger \mathcal{B})$$

For finite-dimensional lattice models (like a spin chain), the infinite-temperature trace is perfectly fine. However, the infinite-temperature trace diverges in a QFT (because the Hilbert space is infinite-dimensional and high-energy UV states ruin the trace). To fix this, we have to evaluate the operator growth at a finite temperature ($\beta > 0$), which physically corresponds to studying operator growth in the background of an AdS black hole. To do this, we use the Wightman inner product:

$$(\mathcal{A}|\mathcal{B})_\beta = \frac{1}{Z(\beta)} \text{Tr} \left(e^{-\beta H/2} \mathcal{A}^\dagger e^{-\beta H/2} \mathcal{B} \right)$$

By splitting the thermal Boltzmann weight $e^{-\beta H}$ in half and wrapping it around the operators, we naturally suppress the high-energy UV divergences. This makes the inner product finite and well-defined for QFTs.

Now that we have a space and defined an inner product, all we need to ensure is that its basis is orthogonal.

3.2 Orthogonality check part 1

To check the orthogonality of our current nested operators $\mathcal{L}^n \mathcal{O}$, we need to ensure that their inner product vanishes. If we take the initial operator $\mathcal{O}$ and the first step $\mathcal{L}\mathcal{O} = [H, \mathcal{O}]$, they are actually orthogonal.

$$(\mathcal{O}|\mathcal{L}\mathcal{O}) = \text{Tr}(\mathcal{O}[H, \mathcal{O}]) = \text{Tr}(\mathcal{O}H\mathcal{O} - \mathcal{O}^2 H)$$

and because the trace is cyclic

$$\text{Tr}(\mathcal{O}H\mathcal{O} - \mathcal{O}^2 H) = \text{Tr}(\mathcal{O}H\mathcal{O} - \mathcal{O}H\mathcal{O}) = 0$$

However, the story is not quiet true for

$$(\mathcal{O}|\mathcal{L}^2 \mathcal{O})$$

. As our Liouvillian superoperator $\mathcal{L}$ generates unitary time evolution ($e^{i\mathcal{L}t}$), it is anti-Hermitian in nature $(\mathcal{A}|\mathcal{L}\mathcal{B}) = -(\mathcal{L}\mathcal{A}|\mathcal{B})$. Hence,

$$(\mathcal{O}|\mathcal{L}^2 \mathcal{O}) = (\mathcal{O}|\mathcal{L}(\mathcal{L}\mathcal{O})) = -(\mathcal{L}\mathcal{O}|\mathcal{L}\mathcal{O})$$

which is just square of its norm, $||\mathcal{L}\mathcal{O}||^2$, and this does not vanishes. This clearly implies our initial raw vectors $\mathcal{L}^n \mathcal{O}$ are not orthogonal.

¹Partition function Z is define as $Z = \text{Tr}(e^{-\beta H})$. As $T \rightarrow \infty$, $Z = \text{Tr}(\mathbb{I}) = D$

3.3 Lanczos Algorithm

We begin with our normalised operator $\mathcal{O}$. This is our first basis vector. Let's call it

$$\mathcal{O}_0 = \mathcal{O}$$

Now we apply the Liouvillian to our operator $\mathcal{L}\mathcal{O}_0$. This new vector is orthogonal to our original operator as we have seen. We defined a new orthonormal basis after renormalization

$$\mathcal{O}_1 = \frac{1}{b_1} \mathcal{L}\mathcal{O}_0 \quad (15)$$

where b_1 is the norm:

$$b_1 = \|\mathcal{L}\mathcal{O}_0\|$$

We apply the Liouvillian to our new vector $\mathcal{L}\mathcal{O}_1$. This vector is not orthogonal to our original operator $\mathcal{O}_0$ hence we define a new vector $|A_2\rangle$ such that

$$|A_2\rangle = \mathcal{L}\mathcal{O}_1 - b_1\mathcal{O}_0 \quad (16)$$

notice that $\mathcal{L}$ is anti-Hermitian hence the projection on $\mathcal{O}_1$ vanishes. Now we can give our new orthonormal basis

$$\mathcal{O}_2 = \frac{1}{b_2} |A_2\rangle = \frac{1}{b_2} (\mathcal{L}\mathcal{O}_1 - b_1\mathcal{O}_0) \quad (17)$$

for our n^{th} orthonormal basis

$$\mathcal{O}_n = \frac{1}{b_n} |A_n\rangle \quad (18)$$

where,

$$b_n = \|A_n\|; \quad |A_n\rangle = \mathcal{L}\mathcal{O}_{n-1} - b_{n-1}\mathcal{O}_{n-2}$$

3.4 orthogonality check part 2

Assume the Lanczos algorithm has worked perfectly up to step n . We have a set of orthonormal basis vectors $\{\mathcal{O}_0, \mathcal{O}_1, \dots, \mathcal{O}_n\}$, meaning $(\mathcal{O}_j|\mathcal{O}_k) = \delta_{jk}$. We want to determine the next vector $\mathcal{O}_{n+1}$. According to the standard Gram-Schmidt procedure, we take our "raw" forward step, $\mathcal{L}\mathcal{O}_n$, and we must subtract its overlap with all previous vectors to make it orthogonal:

$$|A_{n+1}\rangle = \mathcal{L}\mathcal{O}_n - \sum_{k=0}^n c_k \mathcal{O}_k \quad (19)$$

where $c_k = (\mathcal{O}_k|\mathcal{L}\mathcal{O}_n)$. Let's evaluate the coefficients c_k

$$c_k = (\mathcal{O}_k|\mathcal{L}\mathcal{O}_n) = -(\mathcal{L}\mathcal{O}_k|\mathcal{O}_n) \quad (20)$$

as our Liouvillian is anti-Hermitian. Notice, by definition, $\mathcal{L}\mathcal{O}_k$ can just generate the immediate neighbours $\mathcal{O}_{k+1}$ (or $\mathcal{O}_{k-1}$), hence it is a linear combination of $\mathcal{O}_{k+1}$ and $\mathcal{O}_{k-1}$. Now,

For $k < n-1$,

$$c_k = -(\mathcal{L}\mathcal{O}_k|\mathcal{O}_n) = 0 \quad (21)$$

as $\mathcal{L}\mathcal{O}_k$ does not contain any part of $\mathcal{O}_n$ and by assumption they are orthogonal. hence, $c_k = 0$

For $k = n$

$$c_n = (\mathcal{O}_n|\mathcal{L}\mathcal{O}_n) = -(\mathcal{L}\mathcal{O}_n|\mathcal{O}_n) = -c_n \quad (22)$$

which makes $c_n = 0$.

For $k = n-1$

$$c_{n-1} = -(\mathcal{L}\mathcal{O}_{n-1}|\mathcal{O}_n) \quad (23)$$

We know that the forward-stepping part of $\mathcal{L}\mathcal{O}_{n-1}$ was exactly what defined the previous Lanczos coefficient: it steps forward as $b_n\mathcal{O}_n$.

$$c_{n-1} = -(b_n\mathcal{O}_n|\mathcal{O}_n) = -b_n \quad (24)$$

3.5 Krylov Complexity

As we now have a set of orthonormal basis, $\mathcal{O}_n$ which span the entire subspace, we can build our operator at time t as

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} i^n \phi_n(t) \mathcal{O}_n \quad (25)$$

where $\phi_n(t)$ is derive from the inner product

$$\phi_n(t) = i^{-n} (\mathcal{O}_n | \mathcal{O}(t)) \quad (26)$$

Now, Krylov complexity is defined as

$$K(t) = \sum_{n=0}^{\infty} n |\phi_n(t)|^2 \quad (27)$$

Nielsen's complexity is a kinematic quantity. It asks "What is the shortest path between state A and state B" It doesn't care about the actual time evolution of the system.

Krylov complexity is a dynamical quantity. It asks "When the system naturally evolves under its own Hamiltonian, how far down its own intrinsic basis does the operator spread?" There are no arbitrary gate choices, no artificial penalty metrics, and no external architects. The system organically defines its own complexity through its Liouvillian.

3.6 Summary

You start with a simple operator $\mathcal{O}_0$. By repeated application of superoperator, the Liouvillian, $\mathcal{L} = [H, \cdot]$ we build the entire Krylov basis $\mathcal{O}_0, \mathcal{L}\mathcal{O}_0, \mathcal{L}^2\mathcal{O}_0, \dots$. These basis are produced by the Hamiltonian of our system, hence, they are inherent ("internal") to the system. (It does not ask for the external input). You don't worry about discrete vs. continuous; the Liouvillian is perfectly well-defined in QFTs.

Circuit complexity is a kinematic concept, it cares about the shortest path through the entire Hilbert space, regardless of the physical Hamiltonian. Krylov complexity is a dynamical concept, it measures how far the operator has spread, specifically along the one-dimensional trajectory carved out by the system's actual time evolution. By using the Krylov basis, the complexity $K(t)$ becomes an intrinsic, unambiguous property of the specific theory you are studying, which is exactly what we need if we want to extract universal bounds on chaos or match it to a holographic dual.

4 Holography

The realisation that black holes are not truly static. From the outside, a black hole reaches thermal equilibrium very quickly and to a boundary observer, the CFT state looks completely thermalised and unchanging. However, the volume of the Einstein-Rosen bridge continues to grow for an exponentially long time. So the question was "What boundary quantity is still growing, which will be dual to this growing interior volume?" Susskind proposes the dual property as the Quantum Complexity [3, 4]. Later tests on the SYK model revealed the conjecture

$$\dot{C}(t) \propto P \quad (28)$$

where $\dot{C}(t)$ is the rate of complexity and P is the radial momentum of the infalling particle.

4.1 Spread Complexity Rate as Proper Momentum

Caputa et al. (2026) [5] demonstrate a precise relation between the rate of complexity of quantum states excited by local operators in two-dimensional conformal field theories and the radial momentum of particles in three-dimensional anti-de Sitter spacetimes. To achieve this, the authors utilise the spread complexity of states, defined as the average position on the one-dimensional Krylov chain.

For holographic states, the analytical Lanczos coefficients on the Krylov chain are governed by an underlying $SL(2, \mathbb{R})$ symmetry. By identifying a radial coordinate ρ that computes the proper distance to the black hole horizon, the authors established a matching proper radial momentum, P_ρ , for the dual massive particle in the bulk.

The central result of the paper, presented as Equation (29), establishes this precise entry in the holographic dictionary:

$$\dot{C}_K(t) = -\frac{P_\rho(t)}{\epsilon} \quad (29)$$

where the proportionality constant, ϵ , corresponds to the finite energy scale over which the operator is regulated. This equation explicitly equates the rate of growth of the spread complexity in 2D CFTs to the proper radial momentum of dual massive particles in the bulk.

References

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